

Mikayla Newbrey

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Education

University of Massachusetts Amherst

PH.D. STUDENT, ORGANISMAL & EVOLUTIONARY BIOLOGY PROGRAM

Amherst, Massachusetts

2025 - Present

- Advisor: Dr. Lisa Komoroske
- Expected 2031

University of Massachusetts Amherst

BACHELOR OF SCIENCE IN BIOLOGY

Amherst, Massachusetts

2022-2024

Cape Cod Community College

ASSOCIATE OF ARTS IN BIOLOGY

Barnstable, Massachusetts

2019-2022

Work & Research Experience

PhD Research

ENVIRONMENTAL CONSERVATION DEPARTMENT, UNIVERSITY OF MASSACHUSETTS AMHERST

Fieldwork: Gloucester, MA

2025 - Present

- Conduct population genomics and kinship analyses using high-throughput sequencing data
- Develop SNP panels for relatedness and population structure using GT-seq approaches
- Conduct riverscape genomics analyses to identify environmental drivers of population structure and adaptive variation

Lab Manager

MOLECULAR ECOLOGY AND CONSERVATION LAB, UNIVERSITY OF MASSACHUSETTS AMHERST

Amherst, MA

2024-2025

- Directed lab operations, including protocol development, sample preparation, and equipment management
- Coordinated multiple research projects and maintained workflow efficiency across experiments
- Trained and mentored undergraduate and graduate researchers in molecular and genomic techniques

Lead Lab Technician

MOLECULAR ECOLOGY AND CONSERVATION LAB, UNIVERSITY OF MASSACHUSETTS AMHERST

Amherst, MA

2022 - 2025

- Optimized Nanopore sequencing workflows through experimental testing and performance evaluation.
- Executed large-scale genomics projects, including library preparation, sequencing, and downstream analysis.
- Developed and maintained reproducible pipelines for genomic data processing.

Lab Technician

CHENG LAB, UNIVERSITY OF MASSACHUSETTS AMHERST

Amherst, MA

2022 - 2023

- Supported experiments on local adaptation and environmental stress responses in marine organisms.
- Maintained animal systems and conducted daily monitoring under controlled conditions.
- Assisted with common garden experiments and data collection.

CAFE Scholars Internship

MOLECULAR ECOLOGY AND CONSERVATION LAB, UNIVERSITY OF MASSACHUSETTS AMHERST

Amherst, MA

June – Aug 2023

- Conducted population genetics analyses to assess relatedness in marine turtles.
- Validated parent–offspring assignments using genomic markers.
- Performed statistical analyses to support conservation-focused research.

WHOI, CC-CREW Internship

TEPOLT LAB, WOODS HOLE OCEANOGRAPHIC INSTITUTION

Woods Hole, MA

June– Aug 2022

- Designed a 213-SNP panel to resolve population structure in invasive European green crabs.
- Implemented targeted genotyping workflows for large-scale population analysis.
- Integrated transcriptomic data to identify signatures of adaptation.

WHOI, Blue Economy Internship

TEPOLT LAB, WOODS HOLE OCEANOGRAPHIC INSTITUTION

Woods Hole, MA

Jan - June 2022

- Investigated thermal tolerance in invasive European green crabs using respirometry and genomic approaches.
- Analyzed physiological and genetic responses to environmental stress.

Scientific Publications

In Review

Dwane, C., L.M. Komoroske, A.L. Rugila, B.P. Bentley, E. Rawson, E. Clark, G. Nichols, **M. Newbrey**, E. Bucari, C. Yan, J. Barley, R. Horrigan, L. McCarthy, N. Duncan, A.B. Juarez Stucker, A.R. Villeneuve, B.S. Cheng (2025) Adaptation to warm environments with a fast pace of life in a marine predatory snail. Preprint | In revision at Ecological Monographs

In Prep

Venkataraman, Y.R., S.K. Shapiro, **M. Newbrey**, C.K. Tepolt. Metabolomic and lipidomic shifts underpin physiological acclimation to thermal stress in the European green crab (*Carcinus maenas*). Manuscript in preparation.

Teaching & Outreach

Introduction to Ecology

TEACHING ASSISTANT

UMass Amherst

Spring 2025

- Instructor: Tristram Seidler

Supplemental Instructor

CAPE COD COMMUNITY COLLEGE

Barnstable, MA

2020-2022

- Collaborated with faculty to develop course materials, including lecture notes, problem sets, and solution guides, to enhance student understanding of complex topics.
- Planned and facilitated study sessions for General Chemistry, General Biology, and Survey of Biology, reinforcing core concepts and fostering collaborative learning.

Student Tutor

CAPE COD COMMUNITY COLLEGE

Barnstable, MA

2020-2022

- Conducted peer-led review sessions to clarify lecture material, address student questions, and prepare students for exams in General Chemistry, General Biology, and Survey of Biology.
- Delivered one-on-one and group tutoring sessions, tailoring content and teaching approaches to accommodate individual learning styles and academic goals.

MENTORSHIP

Cammi Cooper

UMass Amherst

- Undergraduate, Five College Marine Science Program

2025 - Present

Annalise Kuhlmann

Mount Holyoke College

- Undergraduate, Five College Marine Science Program

Fall 2024

Mary Rutter

UMass Amherst

- Undergraduate, Animal Science Independent Study
- Post-Baccalaureate Research Technician

Jan – July 2024

Mikayla Donovan

UMass Amherst

- Undergraduate, Biology Independent Study
- Lee-SIP Summer Scholar Mentee

Jan – May 2024

Emma Ferreira

UMass Amherst

- Undergraduate, Animal Science-Biotechnology Independent Study

2023

Grants & Fellowships

TOTAL FUNDING RAISED: \$7,389

CNS Graduate Student Fellowship

Fall 2025

UMASS AMHERST - GRADUATE SCHOOL (\$7,389)

Awards

Honorable Mention NSF GRFP

GRADUATE RESEARCH FELLOWSHIP PROGRAM

2026

Placed (3rd) Conference-wide Poster Competition

INTERNATIONAL CONFERENCE ON MARINE BIOINVASIONS

Baltimore, Maryland

2023

Presentations

CONFERENCES

A Genetic Toolkit for Understanding the Rapid Expansion of European Green Crabs in the Northeast Pacific	<i>Baltimore, Maryland</i>
INTERNATIONAL CONFERENCE ON MARINE BIOINVASIONS	2023
Quantifying Relatedness of Green Sea Turtles in the Hawaiian Islands	<i>UMass Amherst</i>
CAFE SCHOLARS PROGRAM POSTER CONFERENCE	2023

Professional Development & Relevant Coursework

Landscape Genomics	<i>Scheduled, Summer 2026</i>
PHYSALIA WORKSHOP	
Applied Ecological Statistics	<i>Spring 2026</i>
DR. MEG GRAHAM MACLEAN, UNIVERSITY OF MASSACHUSETTS AMHERST	
Assembling Genomes with Oxford Nanopore: A Hands-On Guide to Long-Read Sequencing	<i>Spring 2024</i>
PHYSALIA WORKSHOP	
Ecology of Fish	<i>Fall 2023</i>
DR. ANDREW DANYLCHUK, UNIVERSITY OF MASSACHUSETTS AMHERST	

Skills

Software & Tools	Programming & Computing	Wet Lab Techniques	Fieldwork
QGIS	R (RStudio)	GT-Seq	Boat operation and trawling
ArcGIS Pro	Python	RAD-Seq	Davit operation
ImageJ	Bash	Whole Genome Resequencing	Seining
Zotero	SLURM	Nanopore Sequencing	Net tows

Certificates

Scientific Diver	<i>2025 (Active)</i>
AMERICAN ACADEMY OF UNDERWATER SCIENCES	
• Certified in scientific diving, including underwater data collection, species surveys, and equipment deployment under research protocols	

Rescue Diver

2025 (Active)

NATIONAL ASSOCIATION OF UNDERWATER INSTRUCTORS

- Trained in emergency response, diver rescue, and in-water incident management

Open Water Scuba

2024 (Active)

NATIONAL ASSOCIATION OF UNDERWATER INSTRUCTORS

- Certified in foundational scuba diving, navigation, and safety procedures